

HIGH-PERFORMANCE STROKE PREDICTION USING CATEGORICAL BOOSTING AND EXPLAINABLE MACHINE LEARNING MODELS

A Project Report

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CERTIFICATE

This is to certify that the project report entitled “**High-Performance Stroke Prediction Using Categorical Boosting and Explainable Machine Learning Models**” is a bonafide record of work carried out by **I. Karthik (22481A0581), K. Santhosham (22481A0592), K. Uma Shankar (22481A05B1), K. Karthik Babu (22481A05B5)** under the guidance and supervision of **P. S. Ganesh Kumar** in the partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering of Jawaharlal Nehru Technological University Kakinada, Kakinada during the academic year 2025-26.

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LIST OF ABBREVIATIONS

Abbreviation	Explanation
CatBoost	Categorical Boosting
SMOTE	Synthetic Minority Over-sampling Technique
XAI	Explainable Artificial Intelligence
SHAP	SHapely Additive exPlanations
KNN	K Nearest Neighbours
SVM	Support Vector Machine
XGBoost	eXtreme Gradient Boosting
ML	Machine Learning
LIME	Local Interpretable Model-Agnostic Explanations

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ABSTRACT

This study presents an extended and enhanced framework for automated stroke prediction by integrating advanced ensemble learning techniques with explainable artificial intelligence and a secure web-based deployment. To improve predictive reliability, Categorical Boosting (CatBoost) is employed, effectively combining the strengths of multiple base learners. The Synthetic Minority Over-sampling Technique (SMOTE) balances learning and makes it easier to generalize in stroke datasets with a lot of class imbalance. The CatBoost-based ensemble model is 95.58% accurate, which is better than separate machine learning classifiers.

Explainable AI methods look at how global and local features affect model predictions to help doctors make better decisions. We also create a web app using Flask that has secure user authentication for real-time testing and user interaction. This makes the system good enough to be used in healthcare. The proposed end-to-end ensemble and explainable framework demonstrates strong potential for accurate, interpretable, and scalable early stroke risk assessment, supporting timely clinical intervention and improved patient outcomes.

Keywords: Stroke Prediction, Machine Learning, Ensemble Methods, Categorical Boosting, Explainable AI, SHAP, SMOTE, Flask Web Application, User Authentication

CHAPTER 1

INTRODUCTION

1.1 Introduction

Stroke is a serious neurological disorder that can lead to death if immediate medical treatment is not provided. It occurs when the blood supply to the brain is interrupted, preventing oxygen from reaching brain cells and causing severe damage to brain tissues [1]. Stroke is one of the leading causes of mortality and long-term disability worldwide, placing a significant burden on healthcare systems, patients, and caregivers. Beyond its clinical impact, stroke also affects an individual's social participation, psychological well-being, and economic productivity, highlighting the importance of proactive prevention and early risk identification strategies [2].

The global incidence of stroke continues to increase due to factors such as population aging, urbanization, sedentary lifestyles, and the rising prevalence of chronic conditions including hypertension, diabetes, and cardiovascular diseases [3]. These factors have increased the need for accurate, scalable, and cost-effective prediction systems that can identify individuals at high risk before a stroke occurs. Early prediction plays a vital role in enabling timely lifestyle interventions, continuous clinical monitoring, and preventive treatments, which can significantly reduce stroke-related mortality and morbidity.

Clinical research indicates that accurate identification and differentiation of stroke-related conditions are essential for effective treatment planning and improved patient outcomes [4]. Epidemiological studies also show that stroke risk factors vary across populations, age groups, and geographic regions, highlighting the limitations of traditional rule-based or population-specific diagnostic approaches [5]. These findings emphasize the need for data-driven systems capable of learning complex relationships from large and heterogeneous healthcare datasets.

In recent years, machine learning-based stroke prediction systems have gained considerable attention due to their ability to analyze large clinical datasets and uncover hidden relationships among multiple risk factors. However, many existing approaches suffer from several limitations, including severe class imbalance between stroke and non-stroke cases,

limited prediction accuracy, and lack of transparency in model decision-making. These issues reduce their acceptance in clinical environments where reliability, interpretability, and trust are essential.

To address these challenges, this research extends traditional stroke prediction frameworks by incorporating advanced ensemble learning techniques, explainable artificial intelligence methods, and secure web-based deployment. By improving predictive performance, enhancing interpretability, and supporting real-world usability, the proposed system aims to contribute to intelligent and reliable stroke risk prediction systems that assist healthcare professionals in early intervention and improved patient care.

1.2 Objectives of the Project

The primary objective of this project is to develop an intelligent machine learning-based system for early stroke prediction using clinical healthcare data. The system is designed to analyze important health-related risk factors such as age, hypertension, heart disease, body mass index (BMI), smoking status, and average glucose level, which significantly contribute to the occurrence of stroke. By leveraging these features, the model aims to identify individuals at high risk and support early preventive measures.

Another key objective is to improve the reliability and performance of the prediction system by addressing class imbalance in the dataset using the Synthetic Minority Over-sampling Technique (SMOTE). The project also focuses on implementing and comparing multiple machine learning algorithms, including Logistic Regression, Naïve Bayes, K-Nearest Neighbor (KNN), Support Vector Machine (SVM), Random Forest, and XGBoost, in order to identify the most effective model for stroke prediction.

Furthermore, the project aims to enhance prediction accuracy through the use of an advanced ensemble learning approach using the CatBoost algorithm, which is capable of handling categorical data efficiently. In addition, Explainable Artificial Intelligence (XAI) techniques are incorporated to improve transparency and interpretability, allowing users to understand how different features influence prediction outcomes. Finally, the system is deployed using a secure Flask-based web application, enabling real-time stroke risk prediction and improving its practical usability in healthcare environments.

1.3 Problem Statement

Stroke is one of the leading causes of mortality and long-term disability across the globe, posing a major challenge to modern healthcare systems. Early identification of individuals at high risk of stroke is essential for reducing fatality rates and improving patient outcomes. However, traditional diagnostic methods primarily depend on manual clinical evaluation and physician expertise, which may not always be sufficient to detect early warning signs, especially when multiple risk factors are involved.

Healthcare data is inherently complex and multidimensional, consisting of various interrelated attributes such as age, lifestyle habits, medical history, and chronic conditions like hypertension, diabetes, and cardiovascular diseases. These factors often interact in non-linear ways, making it difficult for conventional rule-based or statistical approaches to capture meaningful patterns for accurate prediction. As a result, there is a high possibility of delayed diagnosis or misclassification of stroke risk.

In addition, medical datasets used for stroke prediction often suffer from class imbalance, where the number of stroke cases is significantly lower than non-stroke cases. This imbalance causes machine learning models to become biased toward the majority class, leading to poor detection of actual stroke cases. Another critical issue is the lack of interpretability in many existing machine learning models, which makes it difficult for healthcare professionals to trust and adopt these systems in real-world clinical practice.

Furthermore, many existing stroke prediction systems remain limited to theoretical or experimental stages and are not deployed in practical environments. This lack of accessibility and usability reduces their real-world impact. Therefore, there is a strong need for a robust, accurate, interpretable, and deployable machine learning-based framework that can effectively handle imbalanced datasets, capture complex relationships among risk factors, and provide reliable early stroke risk prediction to support clinical decision-making.

1.4 Scope of the Research

The scope of this research is centered on designing and developing an intelligent stroke prediction system using advanced machine learning techniques applied to structured clinical healthcare data. The study focuses on analyzing multiple patient-related attributes, including demographic, lifestyle, and medical factors such as age, gender, hypertension, heart

disease, body mass index (BMI), average glucose level, work type, residence type, and smoking status. These features are considered essential indicators for assessing stroke risk.

The research includes comprehensive data preprocessing steps to ensure data quality and consistency. This involves handling missing values, encoding categorical variables, and normalizing numerical features to make the dataset suitable for machine learning algorithms. A key aspect of the study is addressing the issue of class imbalance using the Synthetic Minority Over-sampling Technique (SMOTE), which helps improve the model's ability to accurately identify minority class instances, i.e., stroke cases.

The project further explores and compares the performance of multiple machine learning algorithms such as Logistic Regression, Naïve Bayes, K-Nearest Neighbor (KNN), Support Vector Machine (SVM), Random Forest, and XGBoost. Based on comparative analysis, an advanced ensemble learning approach using the CatBoost algorithm is implemented to enhance prediction accuracy and model robustness. CatBoost is particularly effective in handling categorical data and reducing overfitting, making it suitable for healthcare datasets.

In addition, the research integrates Explainable Artificial Intelligence (XAI) techniques to improve the transparency and interpretability of the prediction model. This allows users, especially healthcare professionals, to understand the contribution of each feature in the prediction process, thereby increasing trust in the system.

Finally, the scope extends to the practical deployment of the trained model through a secure Flask-based web application. This enables real-time stroke risk prediction, making the system accessible and usable in real-world healthcare scenarios. The overall research aims to bridge the gap between theoretical machine learning models and practical healthcare applications by providing a scalable, interpretable, and user-friendly prediction system.

CHAPTER 2

LITERATURE REVIEW

The literature review focuses on analyzing existing research works related to stroke prediction, machine learning applications in healthcare, and the role of explainable artificial intelligence (XAI) in medical decision support systems. Stroke is a complex medical condition influenced by multiple risk factors, and several studies have attempted to identify these factors and develop predictive models using both traditional and advanced computational approaches. This section provides a comprehensive overview of previous research contributions, along with their limitations, which form the foundation for the proposed work.

Early studies primarily focused on identifying clinical risk factors associated with stroke. The work presented in [6] highlights hypertension and diabetes mellitus as major contributors to stroke occurrence. The study emphasizes that individuals with uncontrolled blood pressure and diabetes are at a significantly higher risk. However, the approach relies on traditional statistical analysis and lacks automation, limiting its scalability. Similarly, the study in [7] explores genetic, environmental, and lifestyle-related factors influencing stroke risk. Although it provides valuable medical insights, it does not incorporate computational techniques for predictive modeling, thereby limiting its practical applicability.

Several researchers have also investigated patient behavior and response to stroke symptoms. The work in [8] examines how delays in recognizing symptoms and seeking medical assistance can worsen stroke outcomes. Likewise, the systematic review in [9] analyzes public response to stroke symptoms and identifies a lack of awareness as a key issue. While these studies highlight the importance of early detection and timely intervention, they do not provide automated systems for predicting stroke risk in advance.

Clinical diagnostic challenges have also been widely studied. The research in [10] discusses the difficulty of differentiating stroke from other neurological conditions, emphasizing the need for accurate diagnostic tools. However, this study does not explore machine learning-based solutions. These limitations indicate the necessity of intelligent systems that can assist healthcare professionals in accurate and timely decision-making.

With advancements in artificial intelligence, several studies have explored the use of machine learning techniques for stroke prediction. The systematic review in [11] demonstrates that AI-based models significantly improve diagnostic accuracy in ischemic

stroke detection. Similarly, the work in [15] evaluates multiple machine learning models and reports improved performance compared to traditional methods. However, these studies identify key challenges such as class imbalance, lack of interpretability, and limited real-world validation.

The integration of machine learning with healthcare data has shown promising results. The study in [12] utilizes natural language processing and machine learning techniques to detect stroke from electronic health records. While effective, the approach depends heavily on high-quality textual data, which may not always be available. The work in [17] applies data mining techniques for stroke prediction and demonstrates improved accuracy, but it does not address data imbalance issues, which are critical in medical datasets.

Several researchers have also explored advanced diagnostic techniques using specialized data sources. For instance, the study in [13] proposes an electrical impedance tomography-based approach combined with machine learning for stroke diagnosis. Although the system achieves good accuracy, it requires specialized hardware, making it less practical for widespread use. Similarly, the research in [18] introduces an explainable AI model using EEG signals, but the complexity and cost of data acquisition limit its applicability in real-world healthcare environments.

Ensemble learning methods have gained attention due to their ability to improve prediction accuracy. The work in [19] compares multiple machine learning techniques and concludes that ensemble models outperform individual classifiers. The study in [25] further supports this by demonstrating improved accuracy using combined models. However, these approaches often lack sufficient interpretability, which is crucial in healthcare applications.

Explainable Artificial Intelligence (XAI) has emerged as an important area in medical machine learning. The research in [20] proposes an explainable machine learning pipeline for stroke prediction on imbalanced data, highlighting the importance of transparency in decision-making. Similarly, the study in [29] emphasizes that interpretability is essential for building trust in AI systems, especially in high-risk domains like healthcare. Despite these advancements, many existing models still fail to provide clear explanations for their predictions.

Some studies have also focused on scalability and deployment. The work in [27] introduces a distributed machine learning framework using Apache Spark for stroke prediction, demonstrating the ability to handle large datasets. However, the system requires

high computational resources, making it less suitable for lightweight applications. Additionally, research in [21] and [23] shows improved prediction accuracy using machine learning models, but these systems lack real-world deployment and user accessibility.

Recent advancements in deep learning have also influenced medical prediction systems. The studies in [28] and [30], although focused on other diseases, demonstrate the effectiveness of deep learning in healthcare diagnostics. However, these models often act as black boxes, lacking interpretability and transparency, which limits their adoption in clinical practice.

From the above literature, it is evident that machine learning has significantly improved stroke prediction accuracy compared to traditional methods. However, several limitations still exist, including class imbalance in datasets, lack of interpretability, dependency on specialized data, and limited real-world deployment. These gaps highlight the need for a comprehensive system that combines accurate prediction, data balancing, interpretability, and practical usability.

Therefore, the proposed research aims to address these challenges by developing an advanced stroke prediction system that integrates SMOTE for handling class imbalance, ensemble learning techniques such as CatBoost for improved accuracy, Explainable AI methods for interpretability, and a Flask-based web application for real-time deployment. This approach ensures a reliable, scalable, and user-friendly solution for early stroke risk prediction and healthcare decision support.

CHAPTER 3

PROPOSED METHOD

3.1 Methodology

The proposed methodology aims to design and implement a robust, accurate, and interpretable stroke prediction system using machine learning techniques applied to structured clinical healthcare data. Stroke prediction is a complex problem due to the presence of multiple interdependent risk factors and the imbalance inherent in medical datasets. Therefore, a systematic and well-defined pipeline is adopted to ensure reliable model development and real-world applicability.

The overall methodology follows a sequential workflow consisting of data acquisition, preprocessing, feature transformation, model development, evaluation, and deployment. Each stage contributes to improving the predictive performance and generalization capability of the system. The dataset is first collected and analyzed to understand the distribution of features and target variables. Since raw healthcare data often contains missing values, categorical variables, and inconsistencies, preprocessing is performed to enhance data quality and ensure compatibility with machine learning algorithms.

A critical challenge addressed in this research is class imbalance, where stroke cases are significantly fewer than non-stroke cases. This imbalance can lead to biased models that favor the majority class. To mitigate this issue, the Synthetic Minority Over-sampling Technique (SMOTE) is applied to generate synthetic samples for the minority class, thereby improving the model's ability to detect stroke cases.

Following preprocessing, the dataset is partitioned into training, validation, and testing subsets. Multiple machine learning algorithms are then trained to capture patterns associated with stroke risk. These models are evaluated and compared to identify the most effective approach. To further enhance predictive performance, an ensemble learning strategy using the CatBoost algorithm is adopted. Additionally, Explainable Artificial Intelligence (XAI) techniques are incorporated to provide transparency and interpretability, which are crucial in healthcare applications. The final model is deployed through a web-based interface using Flask, enabling real-time prediction and user interaction.

3.1.1 Comprehensive Overview of the Research Design

The research design follows a structured machine learning framework to ensure systematic development and evaluation of the stroke prediction system. The process begins with data acquisition, where a clinically relevant dataset containing patient attributes is obtained. This dataset forms the foundation for the entire study and is carefully analyzed to understand feature distributions and relationships.

The next phase involves data preprocessing, where data cleaning, transformation, and normalization are performed. Missing values are handled using appropriate techniques, and categorical variables are encoded into numerical form. Feature scaling is applied to ensure uniformity across numerical attributes, which improves the performance of distance-based algorithms such as KNN and SVM.

To address the issue of class imbalance, SMOTE is applied to generate synthetic samples for the minority class. This step is essential to prevent model bias and improve the detection of stroke cases. After preprocessing, the dataset is divided into training, validation, and testing sets. The training set is used to build models, the validation set is used for hyperparameter tuning, and the testing set is used for final evaluation.

Multiple machine learning algorithms are implemented and trained using the processed dataset. Each model learns patterns and relationships between input features and stroke occurrence. The performance of these models is compared using standard evaluation metrics. To achieve higher accuracy and robustness, an ensemble learning approach using CatBoost is employed. Finally, Explainable AI techniques are integrated to interpret model predictions, and the system is deployed for real-time usage.

3.1.2 Data Collection

The dataset used in this research consists of 5,110 patient records with 12 attributes representing both demographic and clinical information. These attributes include age, gender, hypertension, heart disease, marital status, work type, residence type, average glucose level, body mass index (BMI), smoking status, and stroke occurrence.

	id	gender	age	hypertension	heart_disease	ever_married	work_type	Residence_type	avg_glucose_level	bmi	smoking_status	stroke
0	9046	Male	67.0	0	1	Yes	Private	Urban	228.69	36.6	formerly smoked	1
1	51676	Female	61.0	0	0	Yes	Self-employed	Rural	202.21	0.0	never smoked	1
2	31112	Male	80.0	0	1	Yes	Private	Rural	105.92	32.5	never smoked	1
3	60182	Female	49.0	0	0	Yes	Private	Urban	171.23	34.4	smokes	1
4	1665	Female	79.0	1	0	Yes	Self-employed	Rural	174.12	24.0	never smoked	1
...	...	...	...	...	...	...	...	...	...	...	...	...
5105	18234	Female	80.0	1	0	Yes	Private	Urban	83.75	0.0	never smoked	0
5106	44873	Female	81.0	0	0	Yes	Self-employed	Urban	125.20	40.0	never smoked	0
5107	19723	Female	35.0	0	0	Yes	Self-employed	Rural	82.99	30.6	never smoked	0
5108	37544	Male	51.0	0	0	Yes	Private	Rural	166.29	25.6	formerly smoked	0
5109	44679	Female	44.0	0	0	Yes	Govt_job	Urban	85.28	26.2	Unknown	0

5110 rows × 12 columns

Fig 3.1.2.1 Sample View of Stroke Prediction Dataset

The stroke attribute is the target variable, where a value of 1 indicates that the patient has experienced a stroke, and 0 indicates no stroke occurrence. The dataset captures a wide range of factors influencing stroke risk, including physiological conditions, lifestyle habits, and demographic characteristics.

The dataset is suitable for machine learning-based prediction as it contains both numerical and categorical features. However, it also presents challenges such as missing values, categorical variables, and class imbalance, which require careful preprocessing before model development.

3.1.3 Data Preprocessing

Data preprocessing is a crucial step in ensuring the quality and reliability of the dataset. Raw healthcare data often contains missing values, inconsistencies, and categorical attributes that cannot be directly processed by machine learning algorithms.

Initially, missing values are identified and handled using appropriate techniques such as mean imputation or removal of incomplete records. This ensures that the dataset is complete and does not introduce errors during model training. Categorical variables such as gender, work type, residence type, and smoking status are converted into numerical form using encoding techniques such as label encoding or one-hot encoding.

Numerical features such as age, BMI, and average glucose level are normalized to ensure uniform scaling. This is particularly important for algorithms that are sensitive to feature magnitude, such as KNN and SVM.

A significant challenge in the dataset is class imbalance, where stroke cases are much fewer than non-stroke cases. To address this issue, SMOTE is applied to generate synthetic samples for the minority class. This improves the model's ability to learn patterns related to stroke cases and enhances overall prediction performance.

3.1.4 Model Selection and Development

The development of an accurate stroke prediction system requires the implementation and comparison of multiple machine learning algorithms. Each algorithm has its strengths and is capable of capturing different types of patterns in the data.

Logistic Regression is used as a baseline model due to its simplicity and interpretability. Naïve Bayes provides probabilistic classification based on feature independence assumptions. K-Nearest Neighbor (KNN) captures local patterns in the data, while Support Vector Machine (SVM) identifies optimal decision boundaries for classification.

Random Forest and XGBoost are ensemble learning techniques that improve prediction performance by combining multiple decision trees. These models are particularly effective in handling complex and non-linear relationships in the dataset.

To further enhance accuracy, the CatBoost algorithm is employed as the final model. CatBoost is a gradient boosting algorithm that efficiently handles categorical features and reduces overfitting. It uses ordered boosting and symmetric trees to improve prediction performance and generalization capability. Due to its ability to handle heterogeneous data and reduce bias, CatBoost is highly suitable for healthcare prediction systems.

3.1.5 Model Testing and Validation

Model testing and validation are essential steps to ensure the reliability and generalization of the prediction system. The dataset is divided into training, validation, and testing sets to prevent overfitting and ensure unbiased evaluation.

The training dataset is used to train machine learning models, while the validation dataset is used for hyperparameter tuning. This process helps optimize model performance and ensures that the model does not memorize the training data.

The testing dataset, which contains unseen data, is used to evaluate the final model. This provides an accurate estimate of how the model will perform in real-world scenarios. Proper validation techniques ensure that the selected model is both accurate and generalizable.

3.1.6 System Architecture

The system architecture represents the complete workflow of the stroke prediction system, illustrating how data flows through different stages to generate predictions.

The architecture begins with the data input layer, where patient data is collected. This data is passed to the preprocessing module, where cleaning, encoding, and normalization are performed. The dataset is then balanced using SMOTE to address class imbalance.

The processed data is fed into the model training module, where multiple machine learning algorithms are trained. The CatBoost model is selected as the best-performing model based on evaluation metrics. The system also includes an Explainable AI module to provide interpretability.

Finally, the model is deployed through a Flask-based web application, allowing users to input patient data and receive real-time stroke risk predictions. This architecture ensures a seamless flow from data input to prediction output and supports practical healthcare applications.

3.2 Implementation

The implementation phase focuses on transforming the proposed methodology into a functional stroke prediction system using Python and various machine learning libraries. This phase involves executing all steps from data preprocessing to model deployment in a systematic and practical manner. The implementation is carried out using tools such as Python, Scikit-learn, Pandas, NumPy, and CatBoost, which provide efficient support for data handling, model training, and evaluation.

Initially, the dataset is loaded into the system and explored to understand its structure, feature distribution, and potential inconsistencies. Data preprocessing techniques are then applied to clean and transform the dataset into a suitable format for machine learning algorithms. This includes handling missing values, encoding categorical variables, and

normalizing numerical features.

To address the issue of class imbalance, the Synthetic Minority Over-sampling Technique (SMOTE) is implemented using appropriate libraries. SMOTE generates synthetic samples for the minority class, ensuring balanced representation of stroke and non-stroke cases in the dataset.

After preprocessing, the dataset is divided into training, validation, and testing sets. Multiple machine learning models are implemented and trained using the training dataset. Hyperparameter tuning is performed using the validation dataset to optimize model performance and prevent overfitting.

The trained models are evaluated using the testing dataset, and their performance is compared using standard evaluation metrics. Based on the results, the CatBoost model is selected as the final model due to its superior accuracy and robustness. The final model is then integrated with a Flask-based web application, allowing users to input patient data and obtain real-time stroke risk predictions.

3.2.1 Model Training and Evaluation

Model training and evaluation are critical steps in the implementation of the stroke prediction system. During this phase, machine learning algorithms are trained to learn patterns and relationships between input features and the target variable (stroke occurrence).

The dataset is divided into three subsets: training, validation, and testing. The training set is used to train the models, the validation set is used to tune hyperparameters and optimize performance, and the testing set is used to evaluate the final model on unseen data.

Multiple machine learning algorithms are implemented, including Logistic Regression, Naïve Bayes, K-Nearest Neighbor (KNN), Support Vector Machine (SVM), Random Forest, and XGBoost. Each model is trained using the same dataset to ensure a fair comparison of performance.

Ensemble learning techniques are also applied, with CatBoost serving as the final model. CatBoost combines multiple weak learners to create a strong predictive model, improving accuracy and generalization. The training process involves adjusting parameters such as learning rate, depth of trees, and number of iterations to achieve optimal performance.

3.2.1.1 *Training Process*

The training process begins with preparing the stroke dataset and dividing it into training, validation, and testing subsets to ensure proper model development and unbiased evaluation. A suitable split ratio is maintained to allow sufficient data for learning, tuning, and testing.

The CatBoost model is initialized with appropriate hyperparameters and trained using the training dataset. During training, the model learns the relationships between clinical features such as age, hypertension, heart disease, body mass index (BMI), average glucose level, and smoking status, and the target variable representing stroke occurrence. The learning process is driven by minimizing a loss function, which measures the difference between predicted and actual outcomes.

To improve model performance, hyperparameter tuning is performed using the validation dataset. Parameters such as the number of trees, tree depth, learning rate, and regularization terms are carefully adjusted to optimize prediction accuracy and prevent overfitting.

The training process continues iteratively until the model achieves stable performance with minimal error and good generalization capability. The final trained CatBoost model is then selected for evaluation and deployment, as it provides the highest prediction accuracy and reliability for stroke risk prediction.

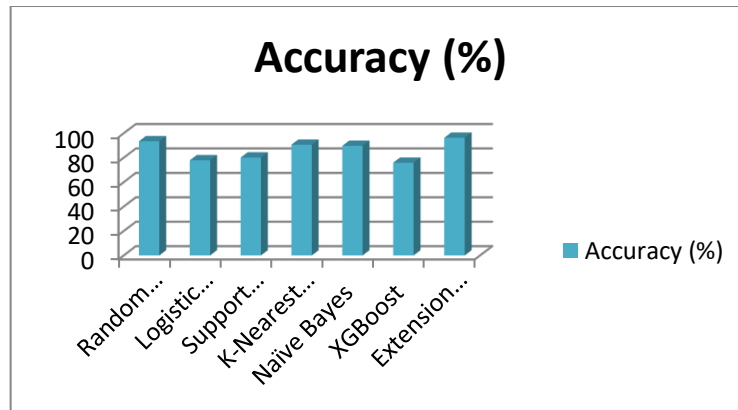
3.2.1.2 *Performance Evaluation Metrics*

After training the models, their performance is evaluated using the testing dataset that was not used during training. This ensures that the evaluation results reflect the model's ability to generalize to unseen data.

Several performance metrics are used for evaluation:

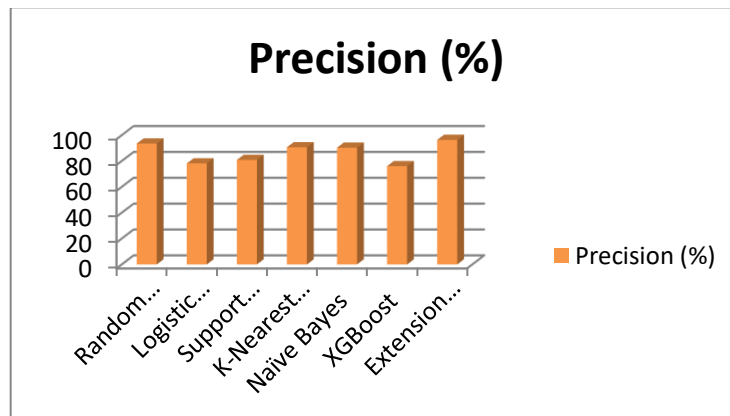
Accuracy: Measures the overall correctness of the model by calculating the proportion of correct predictions. Check the test's reliability by comparing genuine positives and negatives.

$$Accuracy = \frac{(TN + TP)}{T}$$



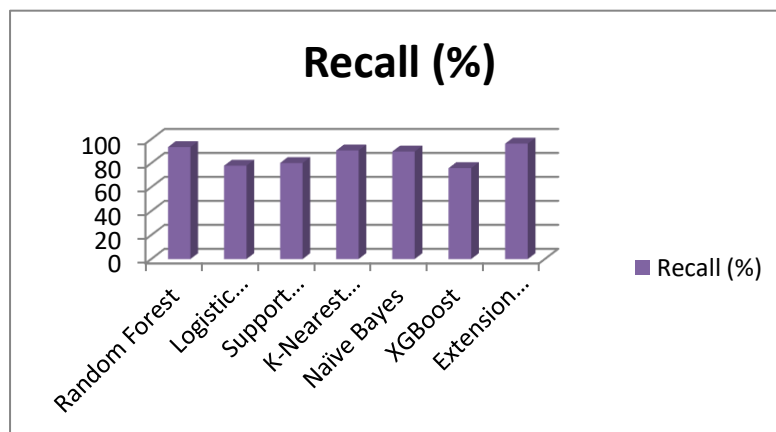
Precision: Measures how many predicted stroke cases are actually correct.

$$Precision = \frac{TP}{(TP + FP)}$$



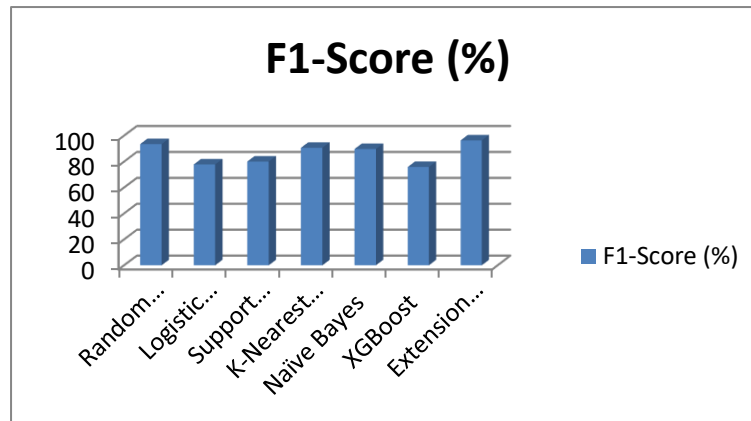
Recall: Measures the ability of the model to identify all actual stroke cases.

$$Recall = \frac{TP}{(FN + TP)}$$



F1-score: Provides a balanced measure of precision and recall and indicates the overall effectiveness of the model.

$$F1 = 2 \cdot \frac{(\text{Recall} \cdot \text{Precision})}{(\text{Recall} + \text{Precision})}$$



These metrics are particularly important in healthcare applications, where correctly identifying stroke cases (high recall) is critical. Based on these evaluation metrics, the CatBoost model demonstrates superior performance compared to other models.

3.2.2 Pseudocode for Model Training and Testing

The pseudocode below represents the complete workflow of the stroke prediction system:

Input: Clinical stroke dataset

Output: Trained stroke prediction model and prediction results

Step 1: Import required libraries (Pandas, NumPy, Scikit-learn, CatBoost)

Step 2: Load the stroke dataset

Step 3: Perform exploratory data analysis

Step 4: Handle missing values

Step 5: Encode categorical variables

Step 6: Normalize numerical features

Step 7: Apply SMOTE to balance the dataset

Step 8: Split dataset into training, validation, and testing sets

Step 9: Initialize machine learning models

- Logistic Regression

- Naïve Bayes

- KNN
- SVM
- Random Forest
- XGBoost

Step 10: Train each model using training dataset

Step 11: Tune hyperparameters using validation dataset

Step 12: Train CatBoost ensemble model

Step 13: Evaluate all models using testing dataset

Step 14: Calculate performance metrics (Accuracy, Precision, Recall, F1-score)

Step 15: Select best-performing model

Step 16: Save trained model

Step 17: Deploy model using Flask web application

Step 18: Accept user input

Step 19: Predict stroke risk

Step 20: Display prediction result

CHAPTER 4

RESULTS AND DISCUSSION

The experimental evaluation of the proposed stroke prediction system was conducted using a clinically relevant dataset consisting of 5,110 patient records with 12 significant health-related attributes. The dataset includes both numerical and categorical features such as age, gender, average glucose level, body mass index (BMI), work type, residence type, smoking status, hypertension, and heart disease. The target variable indicates whether a patient has experienced a stroke. Since the dataset contains a significantly smaller number of stroke cases compared to non-stroke cases, preprocessing techniques along with the Synthetic Minority Over-sampling Technique (SMOTE) were applied to balance the dataset and improve model learning capability.

Multiple machine learning algorithms, including Logistic Regression, Naïve Bayes, K-Nearest Neighbor (KNN), Support Vector Machine (SVM), Random Forest, and XGBoost, were implemented and evaluated using standard performance metrics such as accuracy, precision, recall, and F1-score. The results clearly indicate that ensemble-based methods outperform individual machine learning models in terms of prediction performance. Among all the models, the CatBoost ensemble algorithm achieved the highest accuracy of 95.58%, demonstrating its effectiveness in handling complex clinical data and improving classification performance.

Random Forest and KNN also produced strong results, achieving accuracies above 90%, indicating their capability to capture non-linear relationships within the dataset. Naïve Bayes showed moderate performance due to its assumption of feature independence, while Logistic Regression and XGBoost exhibited comparatively lower accuracy, suggesting limitations in capturing complex feature interactions within the dataset.

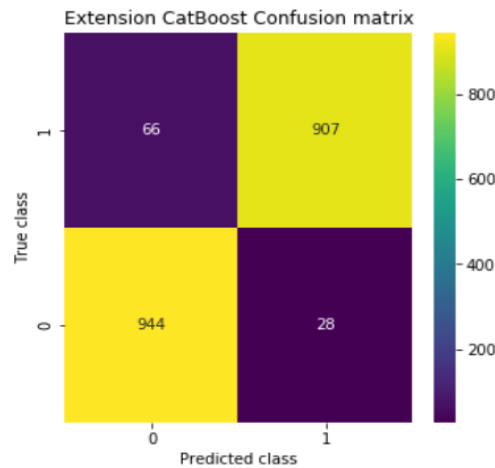


Fig 4.1 Confusion Matrix of the Stroke Prediction Model

The classification performance of the proposed model is further analyzed using a confusion matrix, as shown in Fig. 4.1. The confusion matrix provides a detailed breakdown of prediction results, including true positives, true negatives, false positives, and false negatives.

A higher number of true positives and true negatives indicates that the model accurately identifies both stroke and non-stroke cases. Minimizing false negatives is particularly important in healthcare applications, as failing to detect a stroke case can lead to severe consequences. The confusion matrix demonstrates that the CatBoost model achieves a strong balance between sensitivity and specificity, making it reliable for medical decision support.

Performance Evaluation Metrics

The performance of the prediction models is evaluated using standard metrics:

- **Accuracy** measures the overall correctness of the model.
- **Precision** indicates how many predicted stroke cases are actually correct.
- **Recall** represents the model's ability to identify all actual stroke cases.
- **F1-score** provides a balance between precision and recall.

These metrics collectively provide a comprehensive evaluation of model performance, especially in imbalanced datasets where relying solely on accuracy may be misleading.

Algorithm Name	Precision (%)	Recall (%)	F1-Score (%)	Accuracy (%)
Random Forest	93.53	93.47	93.47	93.47
Logistic Regression	78.19	77.89	77.83	77.89
Support Vector Machine (SVM)	80.87	80.15	80.04	80.15
K-Nearest Neighbor (KNN)	90.64	90.64	90.64	90.64
Naïve Bayes	90.31	89.71	89.68	89.72
XGBoost	76.06	75.88	75.85	75.89
Extension CatBoost	95.61	95.59	95.58	95.58

The comparative results clearly show that the CatBoost model outperforms all other algorithms across all evaluation metrics. Its ability to handle categorical features effectively and reduce overfitting contributes to its superior performance.

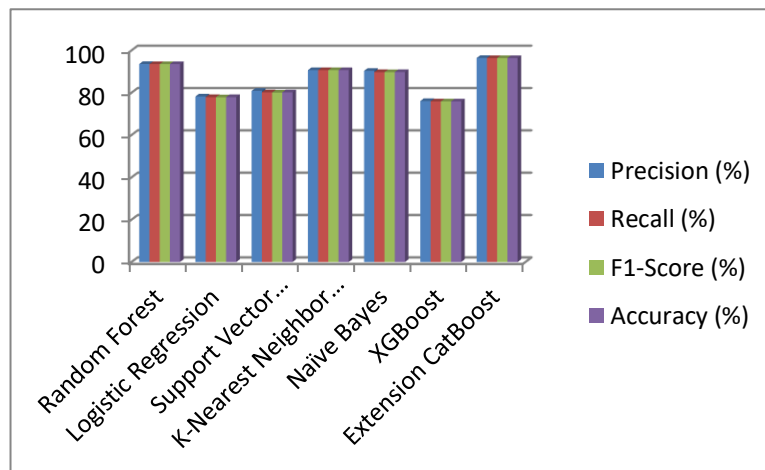


Fig 4.2 Performance Comparison of Machine Learning Algorithms

The graphical representation in Fig. 4.2 highlights the comparative accuracy of different machine learning models. It clearly shows that the CatBoost model achieves the highest accuracy, followed by Random Forest and KNN. This visualization helps in understanding the relative performance of each algorithm more effectively.

The experimental results demonstrate that ensemble learning techniques, particularly CatBoost, are highly effective in handling complex and imbalanced healthcare datasets. The improved performance of CatBoost can be attributed to its ability to manage categorical variables efficiently and reduce overfitting through ordered boosting.

The integration of SMOTE significantly enhances model performance by ensuring balanced representation of stroke cases, thereby improving recall and reducing bias toward the majority class. This is particularly important in medical applications, where correctly identifying positive cases is critical.

Furthermore, the incorporation of Explainable Artificial Intelligence (XAI) techniques improves the interpretability of the model by identifying the most influential features affecting stroke prediction. This enhances trust and usability in real-world clinical environments.

The deployment of the trained model through a Flask-based web application demonstrates the practical applicability of the system. It enables real-time stroke risk prediction, making the system useful for healthcare professionals and supporting early intervention and preventive care.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

This research presents a robust and efficient machine learning-based framework for early stroke prediction using structured clinical healthcare data. The study systematically analyzes various health-related risk factors, including age, hypertension, heart disease, body mass index (BMI), average glucose level, and smoking status, to identify patterns associated with stroke occurrence.

Multiple machine learning algorithms, such as Logistic Regression, Naïve Bayes, K-Nearest Neighbor (KNN), Support Vector Machine (SVM), Random Forest, and XGBoost, were implemented and evaluated to determine their effectiveness in stroke prediction. Among these, the CatBoost ensemble algorithm was selected as the final model due to its superior performance, ability to handle categorical data efficiently, and reduced overfitting.

Data preprocessing played a crucial role in improving model performance. Techniques such as handling missing values, encoding categorical features, and balancing the dataset using the Synthetic Minority Over-sampling Technique (SMOTE) ensured high-quality input data and enhanced the learning capability of the models. The experimental results demonstrate that the CatBoost model achieved the highest prediction accuracy of 95.58%, outperforming all other algorithms.

In addition to accuracy, the integration of Explainable Artificial Intelligence (XAI) techniques significantly improved the transparency and interpretability of the model by identifying the most influential features contributing to stroke prediction. This aspect is particularly important in healthcare applications, where understanding the reasoning behind predictions is essential for building trust among medical professionals.

Furthermore, the deployment of the trained model through a secure Flask-based web application demonstrates the practical applicability of the proposed system. The system enables real-time stroke risk prediction, making it a valuable tool for early detection, preventive healthcare, and clinical decision support.

Overall, the proposed framework provides an accurate, interpretable, and scalable solution for stroke prediction. It bridges the gap between machine learning research and real-

world healthcare applications by combining high predictive performance with usability and transparency.

5.2 Future Scope

Although the proposed system achieves high prediction accuracy and demonstrates strong performance, there are several opportunities for further enhancement and expansion. Future work can focus on improving both the technical capabilities and practical applicability of the system.

One potential improvement is the use of larger and more diverse healthcare datasets collected from different geographical regions and populations. This would enhance the generalization capability of the model and ensure its robustness across varied clinical conditions. Incorporating longitudinal patient data can also improve prediction accuracy by capturing temporal patterns in health conditions.

Advanced deep learning techniques, such as neural networks and hybrid models, can be explored to capture more complex and non-linear relationships within medical data. These models may further improve prediction performance, especially when large-scale datasets are available.

Another important direction is the integration of real-time healthcare data from wearable devices and electronic health records (EHRs). This would enable continuous monitoring of patient health and provide dynamic stroke risk prediction, allowing for timely medical intervention and personalized healthcare recommendations.

The current Flask-based web application can be further enhanced into a comprehensive healthcare decision-support system. This may include features such as mobile application integration, user-friendly dashboards, secure data storage, and role-based access for healthcare professionals. Additionally, incorporating alert systems and personalized recommendations can improve usability and patient engagement.

Future research can also focus on enhancing explainability by integrating advanced XAI techniques that provide more detailed and visual explanations of model predictions. This will further improve trust and adoption of AI-based systems in clinical practice.

In conclusion, the proposed system provides a strong foundation for intelligent stroke prediction, and with further advancements, it has the potential to evolve into a comprehensive, real-time, and clinically reliable healthcare support system.

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(An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)

Seshadri Rao Knowledge Village, Gudlavalleru

Department of Computer Science and Engineering

Program Outcomes (POs)

Engineering Graduates will be able to:

- 1. Engineering knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
- 2. Problem analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
- 3. Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
- 4. Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions, component, or software to meet the desired needs.
- 5. Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
- 6. The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
- 7. Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice. Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

- 8. Individual and Collaborative Team Work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- 9. Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.
- 10. Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
- 11. Life-Long Learning:** Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

Program Specific Outcomes (PSOs)

PSO 1: Design, develop, test and maintain reliable software systems and intelligent systems.

PSO 2: Design and develop web sites, web apps and mobile apps.

PROJECT PROFORMA

PROJECT COs – POs/PSOs MAPPING

Course Title: B. Tech Capstone Project

Course Code: _____

Department: _____

Batch Number: _____

Academic Year: _____

PROJECT PROFORMA

Classification of Project	Application	Product	Research	Review
	√			

1. Project Course Outcomes (COs)

CO No	Course Outcome Statement
CO1	Identify and analyze the problem statement using prior technical knowledge in the domain of interest.
CO2	Design and develop engineering solutions to complex problems by employing systematic approach
CO3	Examine ethical, environmental, legal and security issues during project implementation.
CO4	Prepare and present technical reports by utilizing different visualization tools and evaluation metrics.

2. Program Outcomes (POs)

PO No	Program Outcome Statement
PO1	Engineering Knowledge
PO2	Problem Analysis
PO3	Design / Development of Solutions
PO4	Conduct Investigations of Complex Problems
PO5	Engineering Tool Usage
PO6	The Engineer and the World
PO7	Ethics
PO8	Individual and Collaborative Team Work
PO9	Communication
PO10	Project Management and Finance
PO11	Life Long Learning

3. Program Specific Outcomes (PSOs)

PSO No	PSO Statement
PSO1	Design, develop, test and maintain reliable software systems and intelligent systems.
PSO2	Design and develop web sites, web apps and mobile apps

4. CO – PO / PSO Mapping Table (3–High, 2–Moderate, 1–Low)

CS3528: MAIN PROJECT														
Course Outcomes	Program Outcomes and Program Specific Outcome													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11		PSO 1	PSO 2
CO1	3	3	1				2	2	2				1	1
CO2	3	3	3	3	3		2	2	2		1		3	3
CO3	2	2	3	2	2	3	3	2	2	2			3	
CO4	2	2	2		3			3	3	2	2		2	2

Justification of CO-PO/PSO Mapping

CO1: Identify and analyze the problem statement

This outcome strongly maps to **PO1** and **PO2** as students apply fundamental and domain knowledge to understand and define complex problems. It has a low mapping with **PO3** since design activities are minimal at this stage. Moderate mapping is observed with **PO7**, **PO8**, and **PO9** due to ethical awareness, collaborative discussions, and effective communication during problem identification. It has slight alignment with **PSO1** and **PSO2**, as implementation aspects are not yet emphasized.

CO2: Design and develop engineering solutions

This outcome strongly maps to **PO1**, **PO2**, and **PO3** as it involves applying knowledge, analyzing problems, and designing appropriate solutions. It also strongly aligns with **PO4** through validation and experimentation, and **PO5** via the use of modern tools and technologies. Moderate mapping is observed with **PO7**, **PO8**, and **PO9** as development requires collaboration and adherence to professional practices. A slight mapping exists with **PO11** as students explore new tools and technologies. This outcome strongly supports **PSO1** and **PSO2**, as it focuses on developing reliable software systems and applications.

CO3: Examine ethical, environmental, legal, and security issues

This outcome moderately maps to **PO1** and **PO2** as it involves understanding and analyzing contextual issues. It strongly aligns with **PO3** since solutions must consider constraints such as safety, sustainability, and societal impact. Moderate mapping is observed with **PO4** and **PO5** in evaluating these aspects using appropriate tools and methods. It strongly maps to **PO6** and **PO7** due to emphasis on societal, environmental, legal, and ethical responsibilities. It also moderately contributes to **PO8**, **PO9**, and **PO10** through awareness and discussion of these issues. Strong alignment is observed with **PSO1**, as secure and reliable system development requires these considerations.

CO4: Prepare and present technical reports

This outcome moderately maps to **PO1**, **PO2**, and **PO3** as it involves documentation of knowledge, analysis, and design aspects. It strongly aligns with **PO5** through the use of visualization and reporting tools. Strong mapping is observed with **PO8** and **PO9** as effective collaboration and presentation skills are essential. It moderately contributes to **PO10** through structured organization and reporting of project work. It also shows moderate alignment with **PO11** as students enhance their ability to learn and present new concepts. This outcome moderately supports **PSO1** and **PSO2**, since documentation and presentation are integral parts of software and application development processes.

Note: Map each project outcomes with Pos and PSOs with either 1 or 2 or 3 based on level of mapping as follows:

1-Slightly (Low) mapped 2-Moderately (Medium) mapped 3-Substantially (High) mapped.

5. Sustainable Development Goals (SDG) Alignment

Project Title: High-Performance Stroke Prediction Using Categorical Boosting and Explainable Machine Learning Models

SDG No	Goal	Relevance to the Project
SDG 3	Good Health and Well-being	Focuses on early stroke prediction, enabling timely medical intervention and reducing mortality and long-term disability.
SDG 9	Industry, Innovation & Infrastructure	Applies advanced machine learning models and deploys a web-based system, promoting innovation in healthcare technology.
SDG 10	Reduced Inequalities	Provides an accessible and scalable solution that can support healthcare services in low-resource and underserved areas.
SDG 17	Partnerships for the Goals	Encourages collaboration between healthcare institutions, researchers, and technology platforms to improve stroke prediction systems and share medical insights for better global health outcomes.

Signature(s) of Students:

Signature of Project Supervisor:

- 1.
- 2.
- 3.
- 4.

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ICCET 2026" held during 22nd & 23rd March 2026 at Prince Shri Venkateshwara
Padmavathy Engineering College, Chennai, India.


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14th INTERNATIONAL CONFERENCE ON CONTEMPORARY ENGINEERING AND TECHNOLOGY 2026



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ORGANIZATION OF SCIENCE & INNOVATIVE ENGINEERING AND TECHNOLOGY (OSIET), CHENNAI, INDIA.

IN COLLABORATION WITH
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**PRINCE SHRI VENKATESHWARA PADMAVATHY ENGINEERING COLLEGE
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Has presented a paper titled
**High-Performance Stroke Prediction Using Categorical Boosting and Explainable Machine
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